

Causal Machine Learning: Homework 2

Due date: October 6, 11:59PM

Your name:

```
library(hdm)
library(keras3)
library(dplyr)
library(sandwich)
```

Potential Outcome Framework

1. In lecture, we showed that the observed difference in means (Average Predictive Effect, APE)

$$\pi = E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]$$

can be decomposed into the Average Treatment Effect on the Treated (ATT) plus a selection bias term. Show that π can equivalently be decomposed into the Average Treatment Effect on the Controls (ATC) plus a selection bias term. In words, explain the meaning of this alternative selection bias.

2. Suppose treatment status D_i is independent of potential outcomes $Y_i(d)$, $d \in \{0, 1\}$ conditional on a set of covariates X_i . Define the Conditional ATE (CATE) as

$$\delta(X_i) = E[Y_i(1) \mid X_i] - E[Y_i(0) \mid X_i]$$

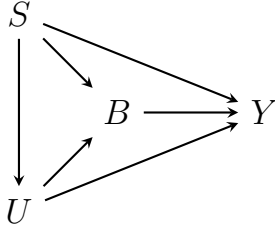
Show that

$$\delta(X_i) = E \left[Y_i \left(\frac{\mathbf{1}(D_i = 1)}{P(D_i = 1 \mid X_i)} - \frac{\mathbf{1}(D_i = 0)}{1 - P(D_i = 1 \mid X_i)} \right) \middle| X_i \right]$$

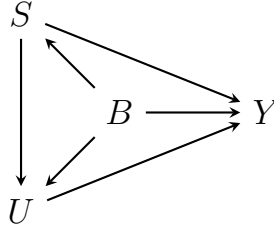
where $P(D_i = 1 \mid X_i)$ is the probability of having $D_i = 1$ given X_i .

Directed Acyclic Graphs (DAG)

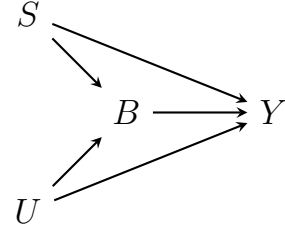
1. The DAGs below graphically describe 3 hypothetical Structural Equation Models (SEMs). state which set of covariates need to be included to identify the causal effect of B on Y .



Case 1



Case 2



Case 3

2. Now focus on Case 3. Suppose the SEM has the following data-generating process

$$Y := S + B + \kappa U + \epsilon_Y$$

$$B := S + U + \epsilon_B$$

$$S := \epsilon_S$$

$$U := \epsilon_U$$

where ϵ_U , ϵ_Y , ϵ_S , and ϵ_B are independent $\mathcal{N}(0,1)$ shocks. Show the following equation holds, and explain the implication of the equation.

$$E[Y | S, B] = (1 - \kappa/2)S + (1 + \kappa/2)B$$

Test of Convergence Hypothesis

In lecture, we provided an empirical example of partialling-out with Double LASSO to estimate the regression coefficient β_1 in the high-dimensional linear regression model. Specifically, we estimated how the rates at which economies of different countries grow (Y_i) are related to the initial wealth levels in each country (D_i) controlling for country's institutional, educational, and other similar characteristics (X_i). The relationship is captured by β_1 , the *speed of convergence/divergence*, which measures the speed at which poor countries catch up ($\beta_1 < 0$) or fall behind ($\beta_1 > 0$) rich countries, after controlling for X_i .

In this assignment, I want you to use the formal procedure of Double Machine Learning to re-estimate β_1 . Specifically, I want you fine-tune a shallow one-layer Neural Network

to estimate the nuisance function for Y_i and D_i using X_i , and residualize Y_i and D_i using cross-fitting.

```
library(hdm)
## function to get data
getdata <- function(...) {
  e <- new.env()
  name <- data(..., envir = e)[1]
  e[[name]]
}

## now load your data calling getdata()
growth <- getdata(GrowthData)

## create the outcome variable y, treatment d, and control covariates x
y <- growth$Outcome
x <- growth[-which(colnames(growth) %in% c("Outcome", "intercept", "gdpsh465"))]
d <- growth$gdpsh465
```

I have already implemented a Neural Network model using `keras3` and performed a manual grid search to fine-tune its hyperparameters. The tuning procedure evaluated each candidate model on a 20% validation split, and the best configuration was chosen based on the lowest validation loss. The default hyperparameter grid includes:

- Number of hidden neurons (`units`): 10, 50, 100
- Learning rate (`lr`): 0.001, 0.02, 0.05
- Batch size: 15 (no tuning)
- Number of epochs: 50, 100, 200, 400

This setup allowed the model to be optimized over different network sizes, learning speeds, mini-batch sizes, and training durations. I also used early stopping based on 10 epoch's validation loss.

```
## build and fit a keras model
build_and_fit <- function(x, y, units, lr, batch_size, epochs) {
  model <- keras_model_sequential() |>
    layer_dense(units = units, activation = "relu", input_shape = c(ncol(x))) |> ## hidden
    layer_dense(units = 1) |>
    compile(
      optimizer = optimizer_adam(learning_rate = lr), ## default optimizer is Adam
```

```

    loss = "mse"
  )

  ## fit the model
  history <- model %>% fit(
    x = as.matrix(x),
    y = y,
    batch_size = batch_size,
    epochs = epochs,
    validation_split = 0.2,
    callbacks = list(
      callback_early_stopping(
        monitor = "val_loss",
        patience = 10, ## early stopping criterion
        restore_best_weights = TRUE
      )
    ),
    verbose = 0
  )

  ## return the loss of the last validation set
  val_loss <- tail(history$metrics$val_loss, 1)

  list(model = model, val_loss = val_loss)
}

## grid search for hyperparameters
tune_keras <- function(x, y,
                      units_grid = c(10, 50, 100),
                      lr_grid = c(0.001, 0.02, 0.05),
                      batch_grid = c(15),
                      epoch_grid = c(50, 100, 200, 400)) {
  results <- expand.grid(units = units_grid,
                        lr = lr_grid,
                        batch_size = batch_grid,
                        epochs = epoch_grid)

```

```

results$val_loss <- NA

for (i in 1:nrow(results)) {
  fit <- build_and_fit(x, y,
                      units = results$units[i],
                      lr = results$lr[i],
                      batch_size = results$batch_size[i],
                      epochs = results$epochs[i])
  results$val_loss[i] <- fit$val_loss
}

## find the best combination of hyperparameter
best <- results[which.min(results$val_loss), ]
return(best)
}

```

Now I want you to fine-tune the NN model *before* cross-fitting. That is, do not fine-tune the model within each $K - 1$ fold, but only fine-tune model once based on the whole sample. Use the default grid search for Y , but use the following grid search for D :

- Number of hidden neurons (`units`): 10, 50, 100
- Learning rate (`lr`): 0.001, 0.02, 0.05
- Batch size: 15 (no tuning)
- Number of epochs: 500, 1000, 2000

Then use the NN model with the combination of hyperparameters that reports the lowest validation loss to estimate the nuisance function. You may find such model by calling

```

ymodel <- build_and_fit(
  x[-I[[b]], ], y[-I[[b]]], ## suppose you are retrieving the nuisance function for
  units = best_y$units,
  lr = best_y$lr,
  batch_size = best_y$batch_size,
  epochs = best_y$epochs
)$model

```

Residualize your Y and D and report the point estimate of β_1 and its standard error using the standard DML pipeline. What can you conclude from this practice? Note this

may take a while for R to complete.

```
## DML using NN - your code here  
dmlNN_for_plm <- function(x, d, y, nfold = 5) {  
  
}  
  
## execute your DML NN function  
set.seed(1)  
dmlNN_for_plm(x, d, y, nfold = 5)
```